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Strategic Planning Analysis and Transformation (SPAT) Request For Proposal

OMG Document: BMI/2014-11-01

Letters of Intent due: August, 2015
Submissions due: February, 2016

Objective of this RFP

Strategic planning is typically a manual process supported by word processing and diagramming tools. BMM (Business Motivation Model) provides a structure for development of strategic plans along with a model for the capture and analysis of strategic plans. However, analysis of the feasibility and impact of strategic plans is limited because available business modelling tools are designed to represent operational detail. These models, such as BPMN (Business Process Model and Notation), CMMN (Case Management Model and Notation) and DMN (Decision Model and Notation) models are too complex to develop rapidly, and too detailed to support executive-level analysis and transformation management. VDML (Value Delivery Modelling Language) supports modelling from a management perspective, but it is not integrated with BMM.

This RFP solicits proposals for the following:

- Provide a linkage of strategic plans of BMM to current and future-state VDML models in order to provide business leaders with an integrated model for capture and analysis of strategic plans.
- Represent current, what-if and future states of the business with VDML models to support consideration of capabilities, value streams, activity networks, value creation and measurements for strategic plans.
- Provide top management with the information they need to define shared business requirements for implementation of strategic plans, to support consistent phases of transformation for use by transformation participants and to set objectives for guidance and accountability for transformation progress.
- Link business objectives and critical success factors to the creation of stakeholder value of current, what-if and future business designs.
- Identify responsible organizations and roles for specific business change requirements at phases of transformation.
- Provide for support of various management views and measurements that currently exist, such as the Balanced Scorecard and Strategy Maps, to assist business leaders in assessing the impact of strategic plans and management of business transformation.

For further details see Section 6 of this document.

6 Specific Requirements on Proposals

This section contains the requirements that are specific to this RFP.

6.1 Problem Statement

Business has become increasingly complex and dynamic. Companies must optimize across lines of business and outsource commodity operations to realize greater economies of scale. Advances in technology frequently make products and business operations obsolete. A complex, competitive environment is created by global markets and competitors, changing economic, social, and political circumstances along with evolving customer expectations and government regulations.

Traditional approaches to strategic planning are no longer sufficient. Word processing and drawing tools do not help manage the complexity of today's business for strategic planning and transformation. The consequences of strategic initiatives are frequently not well understood when a transformation is undertaken. The cost of transformation, the organizational consequences, capability requirements and the impact on customer value are not well understood. Strategic plans are typically too abstract, subject to interpretation so that various aspects of implementation may be uncoordinated and incompatible. As transformation is under way, new strategic initiatives may need to be incorporated into the plan. Development of requirements and planning of transformation efforts require a more detailed and consistent design of the future business from an enterprise perspective along with measurements of progress along the way.

OMG has two specifications in particular that can assist in these efforts: BMM (Business Motivation Model) for strategic planning, and VDML (Value Delivery Modelling Language) for business design and value delivery analysis. However, there is a need to bring these specifications together to provide an integrated, seamless model to support the strategic planning analysis and transformation effort with management views. That integrated modelling environment is the subject of this RFP.

Strategic planning and transformation are no longer an exercises for annual course correction, they must be a continuous process.

6.2 Scope of Proposals Sought

The following paragraphs describe the scope of proposals sought in response to this RFP.

- Integrated business models shall represent (1) strategic plans, (2) transitional states of the business, (3) the relationship between the strategic plan and the transitional states of the business, (4) the relationships between the transitional states and the modelling elements and (5) support for existing views and reports used by business leaders, such as Balanced Scorecards and Strategy Maps.
- The specification is not required to specify the implementation of specific management views, but the RFP response shall demonstrate that the models at least will support well-known, existing techniques. It is expected that this more robust modeling capability will drive the development of new, potentially more interactive views of the current business, strategic plans and transformation initiatives.
- The integration environment shall provide a meta-model that incorporates and, where necessary, extends BMM, VDML and SMM (Structured Metrics Metamodel) such that an integrated model may be created by importing a compliant BMM model and related, compliant VDML and SMM models. The import shall enable the imported BMM, VDML and SMM models to be developed independently during exploratory activities as well as plan development and transformation phases. As a result, SPAT is a composite of BMM, VDML and SMM along with the elements unique to SPAT that integrate and extend these models to provide a seamless modeling environment.
- This specification shall not address integration with BPMN, CMMN, DMN nor other OMG system design specifications because SPAT represents a higher level of abstraction that is appropriate to support top management, and because application of BPMN, CMMN, DMN and other system design tools will require the additional development of operational details and application of technical considerations. A SPAT model effectively becomes a specification of detailed requirements for the implementation of a strategic plan.
- This specification does not require integration of SBVR since integration of SBVR (Semantics of Business Vocabularies and Rules) capabilities—multiple vocabularies, semantics and business rules—should be addressed in a future RFP with the purpose of making these capabilities available to a broader set of OMG specifications.

6.3 Relationship to other OMG Specifications and activities [need to add links to documents]

6.3.1 Relationship to OMG specifications

MOF. Specification metamodels must be represented as MOF metamodels.

XMI. Import and export of BMI, VDML and SMM models must comply with XMI.

BMM. The SPAT specification must incorporate BMM.

VDML. The SPAT specification must incorporate VDML.

SMM. The SPAT specification must incorporate SMM as incorporated in VDML.

SBVR. While SBVR is a BMI specification, it provides an orthogonal modeling capability—capture of business vocabularies, semantics and business rules—that should be generalized for application to multiple OMG specifications.

BPMN. BPMN specifies business processes at a more technical, operational level of detail that is not appropriate for use by business leaders for strategic planning and transformation management. Abstract processes represented in VDML require transformation to introduce detailed technical and operational considerations. Transformation of VDML to BPMN is beyond the scope of this RFP.

CMMN. CMMN specifies adaptive business processes at a more technical, operational level of detail that is not appropriate for use by business leaders for strategic planning and transformation management. Abstract processes represented in VDML require transformation to introduce detailed technical and operational considerations. Transformation of VDML to CMMN is beyond the scope of this RFP.

DMN. DMN specifies business rules for expression of business policies and flow-control at an operational level. Business policies and rules are captured in BMM and their general impact on deliverable flows and value creation are represented in VDML. Transformation to DMN is beyond the scope of this RFP.

6.3.2 Relationship to other OMG documents and activities

- Cummins, Fred A., and Henk de Man, *Analysis of the Relationships between VDML and BPMN*, OMG document number bmi/2013-11-01, November, 2013.
- Cummins, Fred A, *VDML Support for Balanced Scorecard and Strategy Map*, Appendix E of VDML specification [to be posted as separate document], November, 2013.
- Cummins, Fred A., *VDML Support for BMM Strategic Planning*, Appendix E of VDML specification [to be posted as separate document], November, 2013.

6.4 Related non-OMG Activities, Documents and Standards

The following documents are of particular interest regarding current management techniques for documenting, visualizing, specifying and measuring progress of strategic plans and transformation efforts

- LEADing Practice has researched and documented de facto standards for business management reports and documents. Strategic planning and transformation management reports are particularly relevant here for consideration of the completeness of the SPAT metamodel. [reference to be determined]
- The Business Architecture Guild has developed the BA Guild BIZ-BOK describing techniques for business architecture development including associated management reports and documents. [need to determine what can be available to non-Guild members.]

6.5 Mandatory Requirements

6.5.1 MOF-Compliant Meta-model

The submission shall provide a MOF-compliant metamodel that complements and/or extends BMM, VDML and SMM metamodels for integration of BMM, VDML and SMM models.

6.5.2 Scope Determined by Current Business Practices

The scope of the submission shall extend beyond BMM, VDML and SMM to support current strategic planning business practices such as the Balanced Scorecard which includes, for example, objectives for facilities and equipment, market share and customer loyalty.

6.5.3 Modelling of Strategic Planning and Transformation

The submission meta-model shall define the integration and extension of a BMM model for strategic planning, VDML models for phases of business analysis, design, and transformation, and an SMM model for measurements for specification and analysis of a strategic plan, alternative business designs and phases of transformation as well as measurements of value and performance.

6.5.4 Links to Business Objectives and Critical Success Factors

The submission shall provide representation and linking of business objectives (measurements of a desired end state of the business) and critical success factors (measurements of transformation performance that are good indicators of progress) to expected measurements of business performance, value creation, stakeholder value and transformation progress.

6.5.5 Import of Independently Developed BMM, VDML and SMM Models

The submission shall support the ability to import and integrate independently developed BMM, VDML and SMM models (by different teams using different modeling tools). A SPAT model shall consist of a BMM model, VDML models supporting transformation phases, an SMM model supporting measurements and the SPAT modelling elements needed for seamless integration and support for analysis and transformation management.

6.5.6 Identification and Annotation of Changes Associated with Transformation Phases

The submission shall support the ability to identify and annotate changes to VDML model elements requiring effort to change the business from one phase to another (i.e., transformation activities) as well as business changes that may extend beyond those changes reflected in VDML models,

6.5.7 Definition and Annotation of Change Clusters

The submission shall support the definition of clusters of business changes that are interrelated to focus on each a functional change to the business that is the responsibility of a specific organization and/or role.

6.5.8 Identification of Dependencies between VDML Changes

The submission shall support the ability to identify dependencies between changes expressed in the VDML phase models as a basis for transformation planning. A dependency indicates that a change X cannot be effective until change Y has been implemented.

6.5.9 BPMN, CMMN and DMN Out of Scope

This submission shall not address integration with or transformation to BPMN, CMMN, or DMN since these should be reserved for a future RFP.

6.6 Non-mandatory features

6.6.1 Business Leader Views

A submission may include optional, graphical representation of generally recognized business leader views that are supported by the integrated model.

6.6.2 SPAT modelling notation

A submission may include notation for creating and modifying the integration and extensions of BMM, VDML and SMM in order to provide a seamless modelling environment for users that create and maintain the models in support of management users..

6.7 Issues to be discussed

6.7.1 Support for Existing Management Views

A submission shall discuss submission support for existing top management views and terminology as identified by LEADIng Practice

6.7.2 Mapping and Integration with BMM, VDML and SMM

A submission shall discuss the mapping and integration of the proposed meta-model to corresponding BMM, VDML and SMM modelling elements.

6.7.3 Integration Examples

A submission shall discuss examples that illustrate integration of BMM, VDML and SMM.

6.7.4 Upward Compatible Changes to BMM, VDML or SMM

If a submission includes changes to BMM, VDML or SMM specifications, the submission shall discuss any restrictions to upward compatibility of models developed with existing specifications.

6.7.5 Example Application Process

A submission shall discuss an example process that supports management with modelling and analysis of business innovations, strategic planning and transformation of the business through multiple phases.

6.8 Evaluation Criteria

The following aspects of a submission will be considered in addition to the quality of the specification and compliance with requirements identified above:

6.8.1 Consistency of Notation, Terminology and Semantics

Consistency of notation, terminology and semantics with BMM, VDML and SMM as well as current business practices as identified by LEADing Practice research.

6.8.2 Illustration of Application

Use cases and examples that illustrate the application of the meta-model and integration of BMM, VDML and SMM.

6.8.3 Realization of Consensus

The realization of consensus among submitters and other interested parties.

6.9 IPR Mode

Every OMG Member that makes any written Submission in response to this RFP shall provide the Non-Assertion Covenant found in Appendix A of the OMG IPR Policy [IPR].

6.10 RFP Timetable

The timetable for this RFP is given below. Note that the Business Modelling and Integration Domain Task Force or its parent Domain Technical Committee may, in certain circumstances, extend deadlines while the RFP is running, or may elect to have more than one Revised Submission step. The latest timetable can always be found at the *OMG Work In Progress* page at <http://www.omg.org/schedules> under the item identified by the name of this RFP.

Event or Activity	Date
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<i>Letter of Intent (LOI) deadline</i>	<i>August, 2015</i>
<i>Initial Submission deadline</i>	<i>February 2016</i>
<i>Voter registration closes</i>	<i>June, 2016</i>
<i>Initial Submission presentations</i>	<i>March, 2016</i>
<i>Revised Submission deadline</i>	<i>August, 2016</i>
<i>Revised Submission presentations</i>	<i>September, 2016</i>

Appendix A References & Glossary Specific to this RFP

A.1 References Specific to this RFP

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Brodie, L. and Gilb, T., *Values for Value*, AgileRecord, October 2010, <http://www.gilb.com/dl448> .

Cummins, Fred A., *Building the Agile Enterprise with SOA, BPM and MBM*, Morgan Kaufman, 2009.

Cummins, Fred A., Building the Agile Enterprise (blog) includes several posts on VDML, 2011-2013, [Value Delivery Modeling Language: An Update](#) .

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Gordijn, J. and Akkermans, H., *Value based requirements engineering: Exploring innovative e-commerce ideas*. In Requirements Engineering Journal, Vol. 8(2):114-134, 2003, <http://e3value.few.vu.nl/docs/bibtex/pdf/Gordijn2003e3value.pdf>, or a popular version of it: <http://e3value.few.vu.nl/docs/bibtex/pdf/Gordijn2001e3value.pdf>.

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Porter, M. E., *Competitive Advantage: Creating and Sustaining Superior Performance*, The Free Press, New York, 1985.

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A.2 Glossary Specific to this RFP

The following are definitions of terms that are specific to this RFP.

Business Motivation Model (BMM)

Business architecture

Case Management Model and Notation (CMMN)

Domain Model and Notation (DMN)

Integrated model

Management view

Objective

Phase

Plan

Structured Metrics Metamodel (SMM)

Transformation

Value

Value Delivery Modelling Language (VDML)

Appendix B General Reference and Glossary

B.1 General References

The following documents are referenced in this document:

[BCQ] OMG Board of Directors Business Committee Questionnaire,
<http://doc.omg.org/bcq>

[Guide] The OMG Hitchhiker's Guide
<http://doc.omg.org/hh>

[INVENT] Inventory of Files for a Submission/Revision/Finalization
<http://doc.omg.org/inventory>

[IPR] IPR Policy

<http://doc.omg.org/ipr>

[ISO2] ISO/IEC Directives, Part 2 – Rules for the structure and drafting of International Standards

<http://isotc.iso.org/livelink/livelink?func=ll&objId=4230456>

[LOI] OMG RFP Letter of Intent template

<http://doc.omg.org/loi>

[MDAc] MDA Guide

<http://www.omg.org/docs/omg/03-06-01.pdf>

[MOF] Meta Object Facility Specification

<http://www.omg.org/spec/MOF/>

[NS] Naming Service

<http://www.omg.org/spec/NAM>

[OMA] Object Management Architecture

<http://www.omg.org/oma/>

[P&P] Policies and Procedures of the OMG Technical Process

<http://doc.omg.org/pp>

[ISO2] ISO/IEC Directives, Part 2 – Rules for the structure and drafting of International Standards

<http://isotc.iso.org/livelink/livelink?func=ll&objId=4230456>

[RM-ODP]

ISO/IEC 10746

[TEMPL] Specification Template

<http://doc.omg.org/submission-template>

[UML] Unified Modeling Language Specification,

<http://www.omg.org/spec/UML>

[XMI] XML Metadata Interchange Specification,

<http://www.omg.org/spec/XMI>

B.2 General Glossary

Architecture Board (AB) - The OMG plenary that is responsible for ensuring the technical merit and MDA-compliance of RFPs and their submissions.

Board of Directors (BoD) - The OMG body that is responsible for adopting technology.

Letter of Intent (LOI) - A letter submitted to the OMG BoD's Business Committee signed by an officer of an organization signifying its intent to respond to the RFP and confirming the organization's willingness to comply with OMG's terms and conditions, and commercial availability requirements.

Mapping - Specification of a mechanism for transforming the elements of a model conforming to a particular metamodel into elements of another model that conforms to another (possibly the same) metamodel.

Metadata - Data that represents models. For example, a UML model; a CORBA object model expressed in IDL; and a relational database schema expressed using CWM.

Metamodel - A model of models.

Meta Object Facility (MOF) - An OMG standard, closely related to UML, that enables metadata management and language definition.

Model - A formal specification of the function, structure and/or behavior of an application or system.

Model Driven Architecture (MDA) - An approach to IT system specification that separates the specification of functionality from the specification of the implementation of that functionality on a specific technology platform.

Normative – Provisions to which an implementation shall conform to in order to claim compliance with the standard (as opposed to non-normative or informative material, included only to assist in understanding the standard).

Normative Reference – References to documents that contain provisions to which an implementation shall conform to in order to claim compliance with the standard.

Request for Information (RFI) - A general request to industry, academia, and any other interested parties to submit information about a particular technology area to one of the OMG's Technology Committee subgroups.

Request for Proposal (RFP) - A document requesting OMG members to submit proposals to an OMG Technology Committee.

Task Force (TF) - The OMG Technology Committee subgroup responsible for issuing a RFP and evaluating submission(s).

Technology Committee (TC) - The body responsible for recommending technologies for adoption to the BoD. There are two TCs in OMG – the *Platform TC* (PTC) focuses on IT and modeling infrastructure related standards; while the *Domain TC* (DTC) focuses on domain specific standards.

XML Metadata Interchange (XMI) - An OMG standard that facilitates interchange of models via XML documents.